A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Anchovy r-selected – blue whale k-selected **1** r-selected: mature quickly, produce many young, disperse widely, short lives – adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) **1** k-selected: mature late, few young with parental care, long-lived – populations stable near carrying capacity (whales, elephants, rhinos) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly **1** An increase in death rate above the low birth rate causes a population crash **1** Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium) – r-selected species rebound within a few years once exploitation stops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	1 mark for the point below. **1** New population = starting population + births + immigrants - deaths - emigrants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** The greatest rate of exploitation (harvest) possible without causing long-term population decline **1 Equals the surplus production:** highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) **1** At carrying capacity net growth is zero so any harvest reduces the population – harvesting above MSY causes decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock • **sustainable** → meeting present needs without stopping future generations meeting theirs • **carrying capacity** → the largest population the area can support **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 2 points below. **1 In-situ:** conserving species in their natural habitat with the whole community – ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding **1** Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	2
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Is the wild population threatened? **1** Is there a genetically diverse captive population? **1** Is in-situ conservation failing? **1** Is keeping a captive population realistic (habitat size, food, relationships, cost)? **1** Is release likely to be successful now or in future (suitable habitat, threats removed)? **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
02.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Blue whale:** habitat size far too large to provide – food (huge quantities of krill / plankton) cannot be supplied – cost **2 Large blue butterfly:** complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs – requires specific ant species and thyme food plant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed **1** Yield mapping of different parts of a field controls fertiliser application in each area for future crops **1** This maximises yield while avoiding wastage / reducing leaching and nutrient pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive **1** They are mainly available in affluent societies – poorer farmers cannot afford the inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
03.3	1 mark for the point below. **1** Provides an income that allows producers to afford basic rights such as water, education, health care and food **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrates are leached into water bodies used as sources of potable water **1** High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2
03.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Anaerobic digestion by microbes in rice padi fields **1** Livestock intestines (ruminant digestion) **1** Anaerobic decay of manure / slurry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
04.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1 South America:** high loss (5.1-5.5 Mha yr^{-1}) falling by about half by 2020 **1 Africa:** loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest **1 Europe:** negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 South America:** clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export – HEP reservoirs – roads improving access – mining **up to 2 Africa:** growing rural populations clearing land for subsistence farming – fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy) – fodder cutting in dry areas **up to 2 Both:** hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted – urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** They create barriers fragmenting wildlife populations **1** Improved access makes it easier to remove timber and farm produce, so clearance spreads along them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	1 mark for the point below. **1** Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades – slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps **1** Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Fixed mooring buoys so boats do not anchor on coral **1** Education of divers / snorkellers not to touch coral or kick up sediment – bans on spear fishing and souvenir collection **1 Fishing controls:** no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling **1** Protection of turtle nesting beaches – control of coastal development – protection of mangroves and seagrass – artificial reefs (reef balls, sunk ships / oil rigs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** No sunlight so no symbiotic algae – polyps rely only on captured food so have far less energy and grow very slowly **1** Damage from trawls takes hundreds of years to repair – the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Deep-water trawling (also by-catch) – oil and gas exploration in deeper water – ocean acidification **1** Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
07.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) **1** Fish are ectotherms so no energy is used for thermoregulation and water supports their mass – poultry grow fast and are kept warm indoors with movement restricted **1** Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Control of movement (housing / restricted grazing) so less energy is used in muscle activity **1** Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature **1 Species selection:** choose species with low FCRs (poultry, fish) – high-energy easily digested diets – slaughter at the fastest growth stage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Only about 10% of the energy at one trophic level is passed to the next – most is lost as heat in respiration / metabolism or in undigested material **1** Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) **1** Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot **1** Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
09.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Size:** shows whether the population is viable / how urgent action is – sets a baseline to judge success – needed to estimate carrying capacity and sustainable off-take **2 Distribution:** shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors – identifies key areas to protect **2** Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor **2 Age structure:** many young indicates a growing population – few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous – k-selected species need breeding adults protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support • **sustainable** → meeting present needs without stopping future generations meeting theirs • **limiting factor** → the factor in shortest supply that holds the rate or population back **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	8

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition **1** Sub-sample counts in known areas scaled up to the whole reserve – mark-release-recapture (Lincoln index) for smaller species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water – distance from the Sun and the Goldilocks zone – speed of rotation reducing temperature extremes – tilt producing seasons – magnetic field deflecting the solar wind – the Moon stabilising the orbit and producing tides • **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • **Development of life:** Archaea at vents – photosynthesis – oxygen reacting with iron then accumulating – ozone layer allowing colonisation of land – aerobic organisms • **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases	25
	• **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios – understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

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